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SPP analysis for interaction in water on Ti and Si.

▼ SPP single layer model

$$\text{beta} := \frac{\omega_l}{c} \cdot \text{sqrt} \left(\frac{\epsilon_1 \cdot \epsilon_2}{\epsilon_1 + \epsilon_2} \right)$$

$$\frac{\omega_l \sqrt{\frac{\epsilon_1 \epsilon_2}{\epsilon_1 + \epsilon_2}}}{c} \quad (1.1)$$

$$\omega_l := \frac{2 \text{ Pi } c}{\text{lambda}}$$

$$\frac{2 \pi c}{\lambda} \quad (1.2)$$

$$\text{beta}$$

$$\frac{2 \pi \sqrt{\frac{\epsilon_1 \epsilon_2}{\epsilon_1 + \epsilon_2}}}{\lambda} \quad (1.3)$$

$$\text{Lambda} := \frac{2 \text{ Pi}}{\text{Re}(\text{beta})}$$

$$\frac{1}{\Re \left(\frac{\sqrt{\frac{\epsilon_1 \epsilon_2}{\epsilon_1 + \epsilon_2}}}{\lambda} \right)} \quad (1.4)$$

$$\text{lambda} := 800\text{e}-9; c := 3\text{E}8; \omega_p := \text{sqrt} \left(\frac{ne \, ec^2}{me \, meff_{\text{Si}} \epsilon_0} \right); \epsilon_0 := 8.85\text{e}-12; ec$$

$$:= 1.6\text{e}-19; me := 9.1\text{e}-31; meff_{\text{Si}} := 0.18; v_{\text{Si}} := (1.1\text{e}-15)^{-1}$$

$$8.00 \, 10^{-7}$$

$$3. \, 10^8$$

$$\sqrt{\frac{ne \, ec^2}{me \, meff_{\text{Si}} \epsilon_0}}$$

$$8.85 \, 10^{-12}$$

$$1.6 \, 10^{-19}$$

$$9.1 \, 10^{-31}$$

$$\begin{aligned} &0.18 \\ &9.090909091 \cdot 10^{14} \end{aligned} \quad (1.5)$$

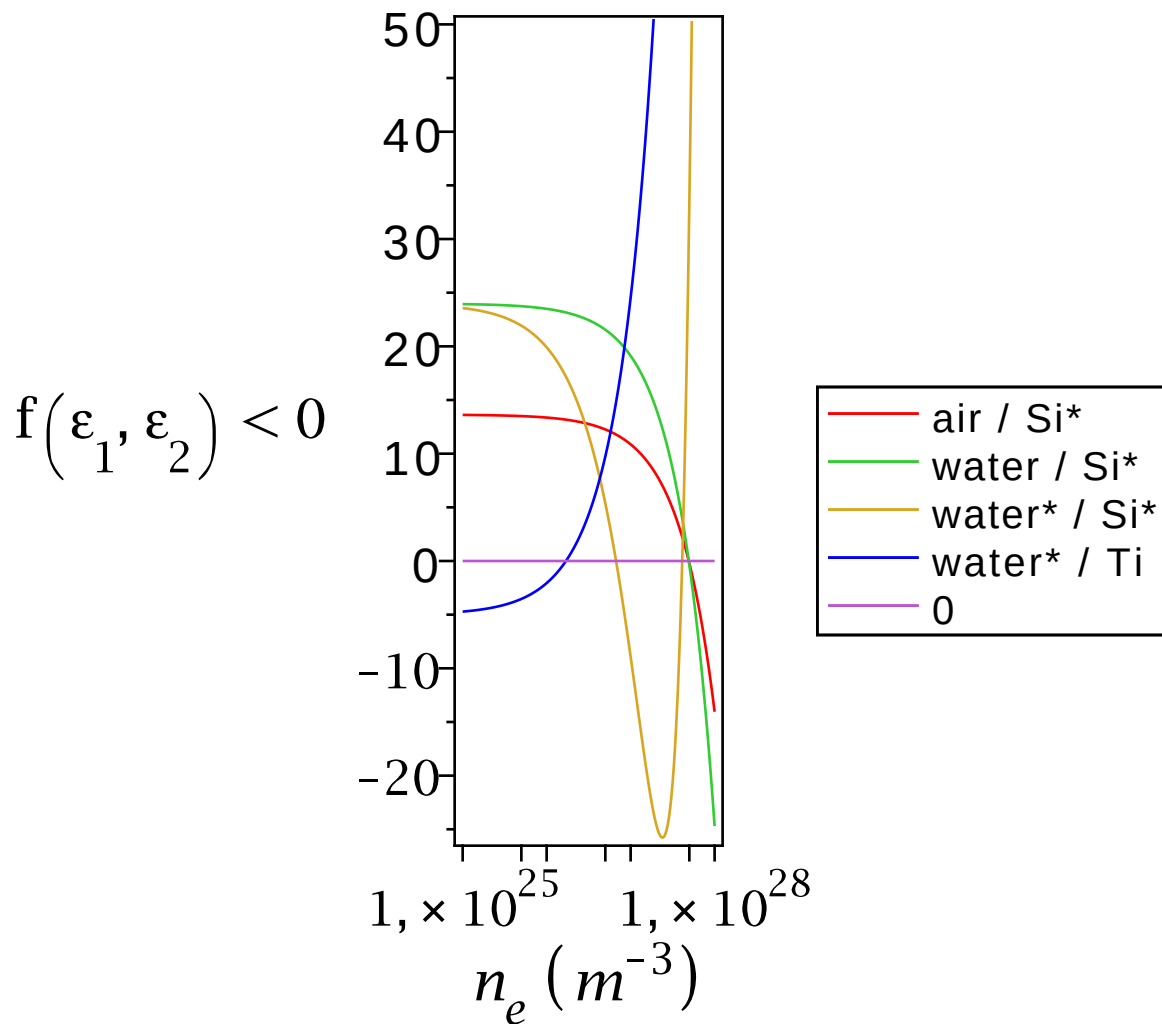
$$\begin{aligned} \epsilon_{Si} &:= \epsilon_{Si0} - \frac{\omega_p^2}{\omega_l^2} \cdot \frac{1}{1 + \frac{I \cdot v_{Si}}{\omega_l}} \\ &\epsilon_{Si0} - \frac{3.139498434 \cdot 10^{-26} ne}{\pi^2 \left(1 + \frac{1.212121212 I}{\pi} \right)} \end{aligned} \quad (1.6)$$

$$\begin{aligned} \epsilon_{Ti} &:= \epsilon_{Tiinf}; \epsilon_{Tiinf} := -2.85 + I \cdot 19.1; \epsilon_{TiO2} := \epsilon_{TiO2inf} - \frac{\omega_p^2}{\omega_l^2} \cdot \frac{1}{1 + I \cdot \frac{v_{TiO2}}{\omega_l}}; \epsilon_{TiO2inf} \\ &:= (2.51975 + I \cdot 0.003)^2; v_{TiO2} := 1e15 \\ &\epsilon_{Tiinf} \\ &-2.85 + 19.1 I \\ &\epsilon_{TiO2inf} - \frac{3.139498434 \cdot 10^{-26} ne}{\pi^2 \left(1 + \frac{1.333333333 \cdot 10^{-15} I v_{TiO2}}{\pi} \right)} \\ &6.349131062 + 0.01511850 I \\ &1. \cdot 10^{15} \end{aligned} \quad (1.7)$$

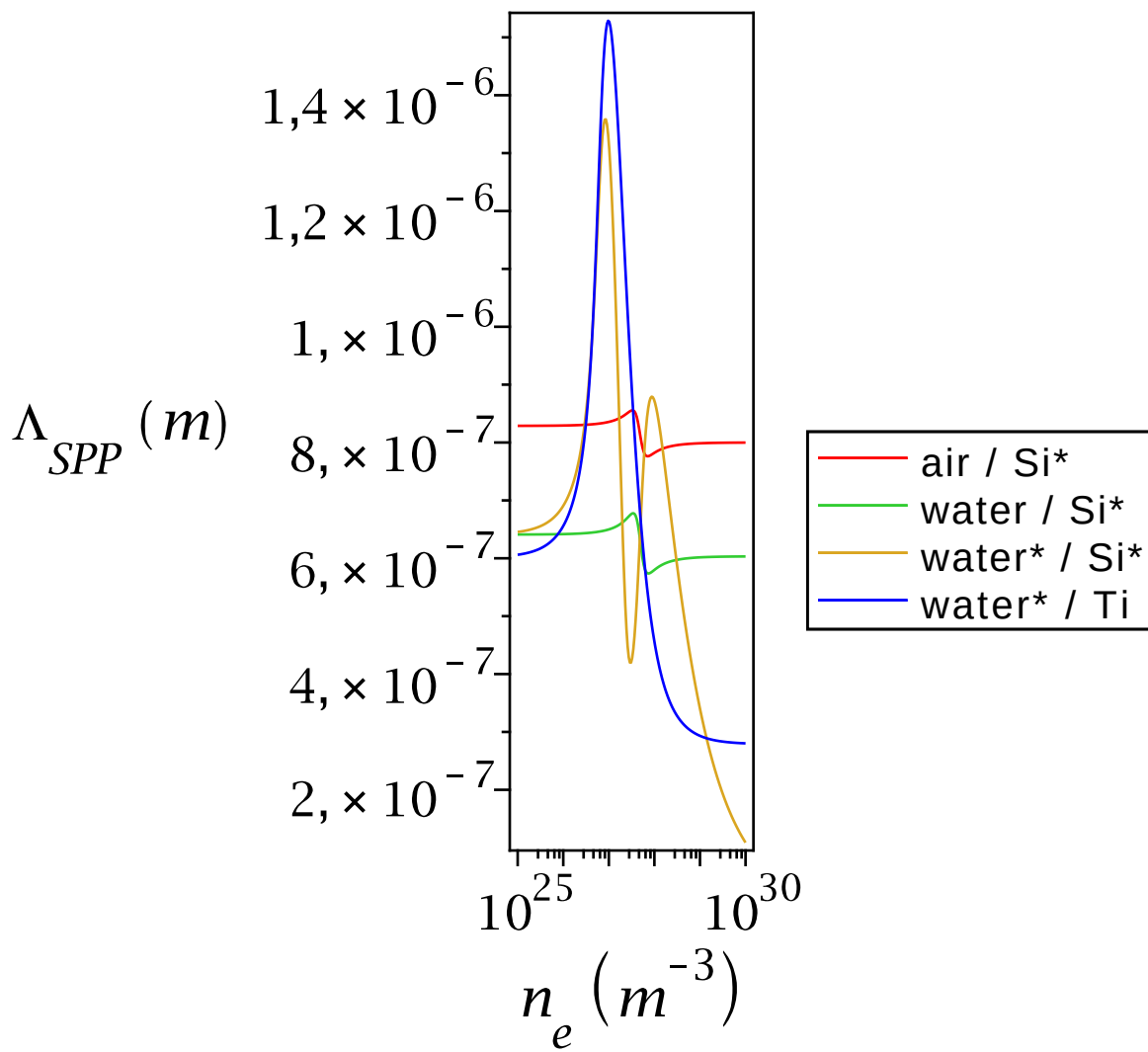
$$\begin{aligned} \epsilon_{air} &:= 1; \epsilon_{Si0} := 13.64 + I \cdot 0.048; \epsilon_{water0} := 1.326^2; \epsilon_{water} := \epsilon_{water0} - \frac{\omega_p^2}{\omega_l^2} \cdot \frac{1}{1 + I \cdot \frac{v_{water}}{\omega_l}}; \\ &v_{water} := 1e15 \\ &1 \\ &13.64 + 0.048 I \\ &1.758276 \\ &1.758276 - \frac{3.139498434 \cdot 10^{-26} ne}{\pi^2 \left(1 + \frac{1.333333333 \cdot 10^{-15} I v_{water}}{\pi} \right)} \\ &1. \cdot 10^{15} \end{aligned} \quad (1.8)$$

with(*plots*) : *semilogplot*([*Re*(ϵ_{air}) · *Re*(ϵ_{Si}) + *Im*(ϵ_{air}) · *Im*(ϵ_{Si}), *Re*(ϵ_{water0}) · *Re*(ϵ_{Si}) + *Im*(ϵ_{water0}) · *Im*(ϵ_{Si}), *Re*(ϵ_{water}) · *Re*(ϵ_{Si}) + *Im*(ϵ_{water}) · *Im*(ϵ_{Si}), *Re*(ϵ_{water}) · *Re*(ϵ_{Ti})

$+ \text{Im}(\epsilon_{\text{water}}) \cdot \text{Im}(\epsilon_{\text{Ti}}), 0]$, $ne = 1\text{E}25..1\text{e}28$, $font = [TIMES, 14, REGULAR]$,
 $numpoints = 200$)



$with(plots) : semilogplot([evalf(eval(eval(Lambda, \epsilon_1 = \epsilon_{\text{air}}), \epsilon_2 = \epsilon_{\text{Si}})),$
 $evalf(eval(eval(Lambda, \epsilon_1 = \epsilon_{\text{water}0}), \epsilon_2 = \epsilon_{\text{Si}})), evalf(eval(eval(Lambda, \epsilon_1$
 $= \epsilon_{\text{water}}), \epsilon_2 = \epsilon_{\text{Si}})), evalf(eval(eval(Lambda, \epsilon_1 = \epsilon_{\text{water}}), \epsilon_2 = \epsilon_{\text{Ti}}))], ne = 1\text{E}25$
 $..1\text{E}30, font = [TIMES, 14, REGULAR], numpoints = 200)$



SPP multilayer model

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$$eq1 := \exp(-4 k_1 a) - \frac{\left(\frac{k_1}{\epsilon_1} + \frac{k_2}{\epsilon_2}\right)}{\frac{k_1}{\epsilon_1} - \frac{k_2}{\epsilon_2}} \cdot \frac{\left(\frac{k_1}{\epsilon_1} + \frac{k_3}{\epsilon_3}\right)}{\frac{k_1}{\epsilon_1} - \frac{k_3}{\epsilon_3}} = 0$$

$$e^{-4 k_1 a} - \frac{\left(\frac{k_1}{\epsilon_1} + \frac{k_2}{\epsilon_2}\right) \left(\frac{k_1}{\epsilon_1} + \frac{k_3}{\epsilon_3}\right)}{\left(\frac{k_1}{\epsilon_1} - \frac{k_2}{\epsilon_2}\right) \left(\frac{k_1}{\epsilon_1} - \frac{k_3}{\epsilon_3}\right)} = 0 \quad (1)$$

$$k_1 := \text{sqrt}((\beta^2 - k_0^2 \epsilon_1)) : k_2 := \text{sqrt}((\beta^2 - k_0^2 \epsilon_2)) : k_3 := \text{sqrt}((\beta^2 - k_0^2 \epsilon_3)) :$$

We investigate now the experimental case we have

$$\begin{aligned} \lambda &:= 800\text{e-}9; c := 3\text{E}8; \omega_{gap} := \sqrt{\frac{ne\,ec^2}{m_e\,m_{effSi}\cdot\epsilon_0}}; \epsilon_0 \\ &:= 8.85\text{e-}12; ec := 1.6\text{e-}19; m_e := 9.1\text{e-}31; m_{effSi} := 0.18; \nu_{Si} := (1.1\text{e-}15)^{-1}; \\ \omega &:= \frac{2\,\pi\,c}{\lambda} \end{aligned}$$

$$\begin{aligned} &8.00\,10^{-7} \\ &3.10^8 \\ &\sqrt{\frac{ne\,ec^2}{m_e\,m_{effSi}\epsilon_0}} \\ &8.85\,10^{-12} \\ &1.6\,10^{-19} \\ &9.1\,10^{-31} \\ &0.18 \\ &9.090909091\,10^{14} \\ &7.500000000\,10^{14}\pi \end{aligned} \tag{2}$$

$$\begin{aligned} \epsilon_{Si} &:= \epsilon_{Si0} - \frac{\omega_{gap}^2}{\omega^2} \cdot \frac{1}{1 + \frac{I\,\nu_{Si}}{\omega}} \\ &3.139498434\,10^{-26} \frac{ne}{\pi^2 \left(1 + \frac{1.212121212\,I}{\pi}\right)} \end{aligned} \tag{3}$$

$$\begin{aligned} \epsilon_{Ti} &:= \epsilon_{TiInf}; \epsilon_{TiInf} := -2.85 + I\,19.1; \epsilon_{TiO2} := \epsilon_{TiO2inf} \\ &- \frac{\omega_{gap}^2}{\omega^2} \frac{1}{1 + I\,\frac{\nu_{TiO2}}{\omega}}; \epsilon_{TiO2inf} := (2.51975 + I\,0.003)^2; \nu_{TiO2} \\ &:= 1\text{e}15; \end{aligned}$$

$$\begin{aligned} &\epsilon_{TiInf} \\ &-2.85 + 19.1\,I \\ &\epsilon_{TiO2inf} - \frac{3.139498434\,10^{-26} \, ne}{\pi^2 \left(1 + \frac{1.333333333\,10^{-15} \, I\,\nu_{TiO2}}{\pi}\right)} \\ &6.349131062 + 0.01511850\,I \\ &1.10^{15} \end{aligned} \tag{4}$$

$$\begin{aligned} \epsilon_{Air} &:= 1; \epsilon_{Si0} := 13.64 + I\,0.048; \epsilon_{Water0} := 1.326^2; \epsilon_{Water} \\ &:= \epsilon_{Water0} - \frac{\omega_{gap}^2}{\omega^2} \frac{1}{1 + I\,\frac{\nu_{Water}}{\omega}}; \nu_{Water} := 1\text{e}15 \end{aligned}$$

$$\begin{aligned}
& 1 \\
& 13.64 + 0.048 I \\
& 1.758276 \\
& 1.758276 - \frac{3.139498434 \cdot 10^{-26} ne}{\pi^2 \left(1 + \frac{1.333333333 \cdot 10^{-15} I nuWater}{\pi} \right)} \\
& 1.10^{15}
\end{aligned} \tag{5}$$

$$\epsilon_1 := \epsilon_{\text{epsilonWater}}, \epsilon_2 := \epsilon_{\text{epsilonWater0}}, \epsilon_3 := \epsilon_{\text{epsilonSi}}, k_0 := \frac{2 \text{ Pi}}{\text{lambda}}$$

$$\begin{aligned}
& 1.758276 - \frac{3.139498434 \cdot 10^{-26} ne}{\pi^2 \left(1 + \frac{1.333333333 I}{\pi} \right)} \\
& 1.758276 \\
& 13.64 + 0.048 I - \frac{3.139498434 \cdot 10^{-26} ne}{\pi^2 \left(1 + \frac{1.212121212 I}{\pi} \right)} \\
& 2.500000000 \cdot 10^6 \pi
\end{aligned} \tag{6}$$

$$\begin{aligned}
& imin := 1 : imax := 10 : neMin := 1e25 : neMax := 1e29 : reason \\
& \quad := \left(\frac{neMax}{neMin} \right)^{\frac{1}{(imax - imin + 1)}} : \\
& jmin := 1 : jmax := 10 : aMin := 0.1e-9 : aMax := 50e-9 : aReason \\
& \quad := \left(\frac{aMax}{aMin} \right)^{\frac{1}{(jmax - jmin + 1)}} :
\end{aligned}$$

Periodicities

Digits := 15

$$15 \tag{7}$$

$$\begin{aligned}
& aList := [seq(aMin \cdot aReason^j, j = jmin..jmax)] \\
& [1.861645567 \cdot 10^{-10}, 3.46572421713075 \cdot 10^{-10}, 6.45195012526601 \cdot 10^{-10}, \\
& \quad 1.20112443492066 \cdot 10^{-9}, 2.23606797968542 \cdot 10^{-9}, 4.16276604189201 \cdot 10^{-9}, \\
& \quad 7.74959494834639 \cdot 10^{-9}, 1.44269990816347 \cdot 10^{-8}, 2.68579588854382 \cdot 10^{-8}, \\
& \quad 5.00000000977443 \cdot 10^{-8}]
\end{aligned} \tag{8}$$

$$\begin{aligned}
& neList := [seq(neMin \cdot reason^i, i = imin..imax)] \\
& [2.511886432 \cdot 10^{25}, 6.30957344726569 \cdot 10^{25}, 1.58489319338942 \cdot 10^{26}, \\
& \quad 3.98107170864403 \cdot 10^{26}, 1.00000000097620 \cdot 10^{27}, 2.51188643445210 \cdot 10^{27}, \\
& \quad 6.30957345342509 \cdot 10^{27}, 1.58489319493659 \cdot 10^{28}, 3.98107171253034 \cdot 10^{28},
\end{aligned} \tag{9}$$

$$1.00000000195240 \cdot 10^{29}]$$

omega; c

$$7.500000000 \cdot 10^{14} \pi$$

$$3 \cdot 10^8$$

(10)

period

period

(11)

$$\text{evalf}(1e9 \cdot 2 \cdot \text{Pi} \cdot \text{abs}(\text{Re}(\text{period}(1e24, 0.3e-9)))^{-1})$$

$$641.033317602294$$

(12)

Digits := 15

$$15$$

(13)

period := proc(neIn, aIn); fsolve((eval(eval(eq1, ne = neIn), a = aIn)), beta, complex);
end proc

proc(neIn, aIn)

(14)

fsolve(eval(eval(eq1, ne = neIn), a = aIn), beta, complex)

end proc

for i from 1 to imax - imin + 1 do for j from 1 to jmax - jmin + 1 do betaSol[i, j]
:= period(neList[i], aList[j]); periodSol[i, j] := evalf(abs(2 · Pi · (Re(betaSol[i,
j]))⁻¹)); DampLength[i, j] := evalf(1e9 · abs(2 Im(betaSol[i, j]))⁻¹); od; od;

Warning, computation interrupted

for i from 1 to imax - imin + 1 do ReEps1[i] := evalf(eval(Re(ε₁), ne = neList[i]));
ReEps2[i] := evalf(eval(Re(ε₂), ne = neList[i])); ReEps3[i] := evalf(eval(Re(ε₃), ne
= neList[i])); od;

Vector(imax - imin + 1, ReEps1)

$$\begin{bmatrix} 1.58820434229638 \\ 0.68519638483979 \\ -5.01239864397314 \\ -40.9617929370067 \\ -267.787136525045 \end{bmatrix}$$

(15)

ReEpsVec1 := Vector(imax - imin + 1, ReEps1); #ReEpsVec2 := Vector(ReEps2);
ReEpsVec3 := Vector(ReEps3);

$$\begin{bmatrix} 1.58820434229638 \\ 0.68519638483979 \\ -5.01239864397314 \\ -40.9617929370067 \\ -267.787136525045 \end{bmatrix}$$

(16)

neVec := Vector(neList)

$$\begin{bmatrix} 6.30957344480193 \cdot 10^{25} \\ 3.98107170553497 \cdot 10^{26} \\ 2.51188643150958 \cdot 10^{27} \\ 1.58489319246111 \cdot 10^{28} \\ 9.99999999999998 \cdot 10^{28} \end{bmatrix} \quad (17)$$

aVec := *Vector(aList)*

$$\begin{bmatrix} 2.51188643150958 \cdot 10^{-10} \\ 6.30957344480193 \cdot 10^{-10} \\ 1.58489319246111 \cdot 10^{-9} \\ 3.98107170553497 \cdot 10^{-9} \\ 1.00000000000000 \cdot 10^{-8} \end{bmatrix} \quad (18)$$

betaVec := *Matrix(periodSol, imax – imin + 1, jmax – jmin + 1)*

DampLengthVec := *Matrix(DampLength, imax – imin + 1, jmax – jmin + 1)*

with(plots) : plot(aVec, [seq(seq(betaVec[i, j], j = jmin..jmin), i = imin..imax)], font = [TIMES, 14, REGULAR])

